

Weight Space Learning for Certifiable Few-shot Transfer Learning

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Abstract

In contemporary deep learning, a prevalent and effective workflow for solving low-data problems is adapting powerful pre-trained foundation models (FMs) to new tasks via parameter-efficient fine-tuning (PEFT). However, while empirically effective, the resulting solutions lack generalisation guarantees to certify their accuracy - which may be required for ethical or legal reasons prior to deployment in high-importance applications. In this paper we develop a novel transfer learning approach that is designed to facilitate non-vacuous learning theoretic generalisation guarantees for downstream tasks, even in the low-shot regime. Specifically, we first use upstream tasks to train a *distribution over PEFT parameters*. We then learn the downstream task by a *sample-and-evaluate* procedure – sampling plausible PEFTs from the trained diffusion model and selecting the one with the highest likelihood on the downstream data. Crucially, this confines our model hypothesis to a *finite* set of PEFT samples. In contrast to the typical continuous hypothesis spaces of neural network weights, this facilitates tighter risk certificates. We instantiate our bound and show non-trivial generalization guarantees compared to existing learning approaches which lead to vacuous bounds in the low-shot regime.

1. Introduction

Generalisation certificates are crucial for high-importance applications where accuracy should be guaranteed for legal or ethical reasons. Guarantees should certify the minimum testing accuracy expected on unseen data drawn from the training distribution. However, it is hard to establish non-trivial guarantees for large neural networks, since large

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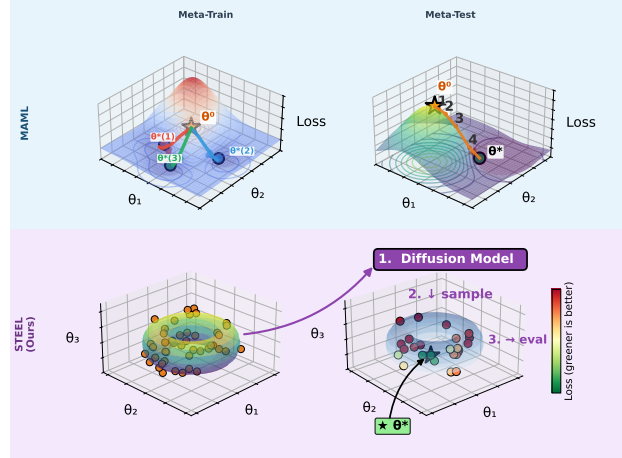


Figure 1. **Contrasting meta-learning paradigms.** *Top row (MAML):* Traditional gradient-based meta-learning. During meta-training, MAML learns an initialization point θ^0 positioned close to all task-specific optima. At meta-test, a few gradient descent steps adapt θ^0 to the new task’s optimum θ^* . θ^0 is the transferred meta-knowledge. *Bottom row (STEEL, Ours):* Manifold-based meta-learning. During meta-training, we collect adapters $\{\theta_i\}$ from upstream tasks and fit a diffusion model to capture their distribution—the task solution manifold. At meta-test, we sample candidates from the learned manifold and select θ^* by evaluating support set loss $r(\theta)$, replacing continuous optimization with discrete selection over a finite hypothesis space. The diffusion model is the transferred meta-knowledge.

learning capacity tends to produce looser guarantees. As such, there have only been a few successful demonstrations of non-vacuous guarantees for contemporary neural networks, even in the large-data regime (Dziugaite and Roy, 2017; Perez-Ortiz et al., 2021; Lotfi et al., 2024).

What about learning with sparse rather than large data? The problem of low-data learning is highly topical, due to the plethora of important limited-data applications (Wang et al., 2020), but challenging due to the difficulty of learning a large number neural network parameters without overfitting. This need has inspired several lines of research that make use of different forms of knowledge transfer, including meta-learning (Hospedales et al., 2021) and parameter-efficient transfer learning (PEFT) (Hu et al., 2021) from foundation models. While PEFT methods have recently been more empirically effective, neither family of approach has produced methods that can provide low-shot generalisation guarantees, to our knowledge. From a learning theoretic perspective this

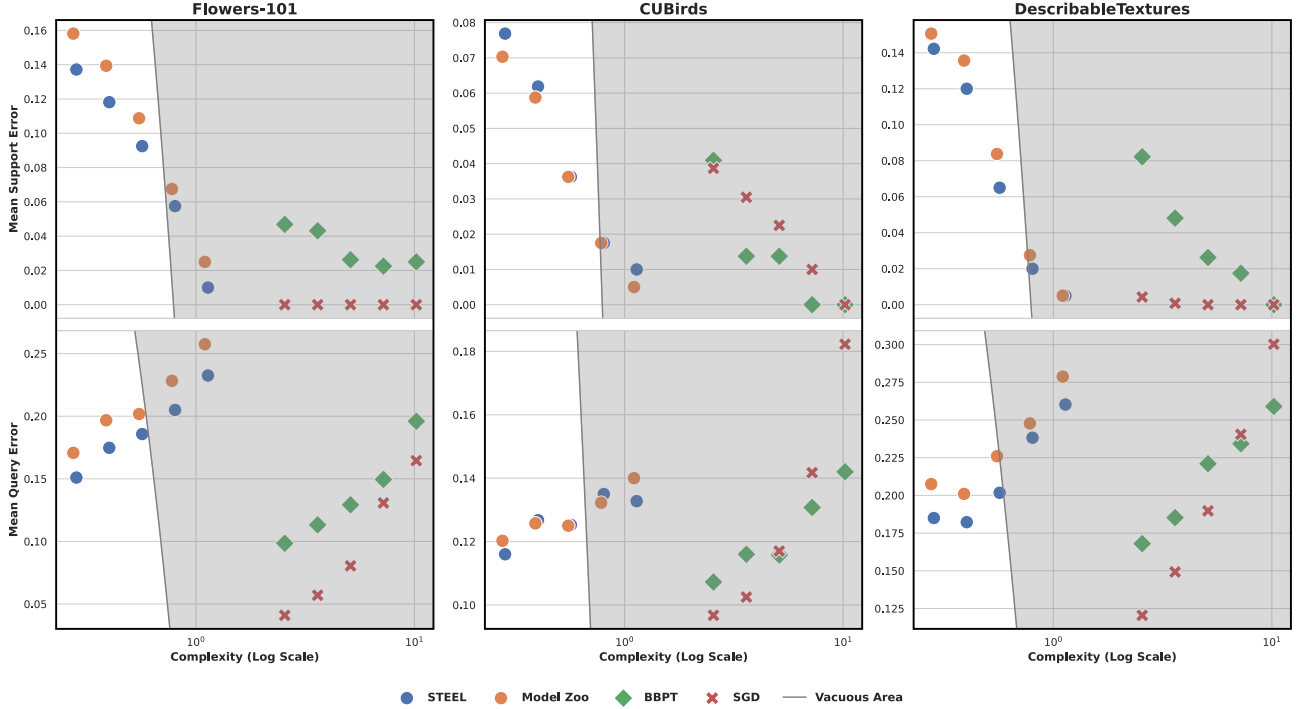


Figure 2. Generalization bounds for adapting CLIP to novel tasks (5-way classification with 1–16 examples per class). Plots show classification error (y-axis) versus the complexity term (x-axis, log scale; square root terms from Equations 3, and 7). Top/Bottom: Mean support/query (train/test) error on new tasks. Shaded regions indicate vacuous bounds, where (support error + complexity) ≥ 1 . Non-vacuous guarantees lie in the unshaded region. Competing methods (SGD, BBPT) fail to achieve non-vacuous bounds. In contrast, our method yields non-vacuous guarantees without significantly compromising training fit (top) or test accuracy (bottom).

is because existing algorithms still search a hypothesis space (e.g., all neural network weights $\theta \in \mathcal{R}^N$) large enough to make known bounds vacuous when instantiated.

This paper introduces a novel approach to knowledge transfer that ultimately learns downstream tasks by *picking from a finite set of hypothesis*, where the set of hypothesis is fit to the upstream tasks. Our method, STEEL (Sample ThEn Evaluate Learner), facilitates using classic finite-hypothesis bounds, which are simple and tight, but not typically used in contemporary machine learning – which focuses on learning continuous value neural network parameters.

More specifically, in the upstream phase, we fit PEFT modules to available source tasks, and then train a parameter diffusion model to generate PEFTs according to this task distribution. In the downstream phase, we learn by sample-then-evaluate instead of traditional gradient descent. PEFT modules, unconditionally generated by the diffusion model, are scored using the target task training set, and learning is to choose the highest scoring module. Compared to the original set of upstream models, the diffusion model can be more compact, and can interpolate between the original model set to achieve higher accuracy. This procedure is gradient-free, which has some scalability benefits (Malladi et al., 2023; Rezk et al., 2024), but more importantly it facilitates the use of PAC-Bayes finite-hypothesis bounds

to provide non-vacuous guarantees, all while maintaining similar empirical accuracy to mainstream few-shot learning approaches. Figure 2 shows some illustrative results, demonstrating our learner’s ability to maintain practical efficacy while being constrained to low enough complexity to provide non-vacuous guarantees (white zone).

Scope. The novelty is methodological. The finite hypothesis class is the structural property that makes certification possible at all—no continuous-parameter method achieves this in the low-shot regime. This positions STEEL as a framework for *certifiable* low-shot transfer: the right tool when deployment requires provable guarantees, complementary to (rather than replacing) unconstrained fine-tuning when raw accuracy is the sole objective. Empirically, STEEL is within a small margin of gradient-based fine-tuning on vision benchmarks (Table 2) and competitive on LLM benchmarks (Table 1), while delivering certificates where every prior method is vacuous.

This opens a concrete research programme—spanning tightening the certificates within the STEEL framework, strengthening empirical accuracy, and extensions to cross-domain transfer. The rest of the paper develops this paradigm: Section 2 reviews the risk-certificate and meta-learning background, Section 3 introduces STEEL and instantiates its PAC-Bayes bound, and Section 5 demonstrates non-vacuous

certificates across LLM and vision benchmarks. Subsequently, we chart what comes next: the open research frontier this paradigm unlocks.

In summary, our contributions are: (1) Introducing a novel learning paradigm for gradient-free transfer learning designed to facilitate accuracy guarantees for downstream tasks, even in the low-shot regime with large foundational model. (2) The first practical demonstration of non-vacuous generalization bounds for low-shot learning in large language and vision architectures.

2. Background

2.1. Risk Certificates for Deep Models

Certifying model generalization performance is fundamental in theoretical machine learning (Vapnik, 1995; Shalev-Shwartz and Ben-David, 2014; Mohri et al., 2018). Vapnik–Chervonenkis (VC), Rademacher, and PAC-Bayes bounds connect empirical risk (computable) to generalization risk (impossible to compute) through inequalities. Here we discuss the core concepts of risk certificates and why they are challenging to apply to deep models.

Let $h \in \mathcal{H}$ be a hypothesis (prediction function $y = h(x)$) and $\mathcal{H}$ the hypothesis space. In deep learning, h corresponds to a model with parameters θ , and $\Theta (\ni \theta)$ represents all possible parameter values, serving as the hypothesis space. A learning algorithm (i.e., SGD) selects θ from Θ given empirical data $S = (x_i, y_i)_{i=1}^n$ sampled i.i.d. from distribution T . The goal is minimizing generalization risk $R(\theta) = \mathbb{E}_{(x,y) \sim T} [l(\theta; x, y)]$, where $l(\theta; x, y)$ is the instance risk. Since computing $R(\theta)$ is impossible without access to T , we minimize empirical risk $r(\theta) = \frac{1}{n} \sum_{(x,y) \in S} l(\theta; x, y)$. Nevertheless, $r(\theta)$ is a surrogate for $R(\theta)$ and does not give any certificate about true generalization risk. Therefore, theoretical bounds relate R and r as follows. For any $\theta \in \Theta$:

$$R(\theta) \leq r(\theta) + \text{ComplexityTerm}(\dim(\Theta), n) \quad (1)$$

where the complexity term depends on data size n and hypothesis space complexity $\dim(\Theta)$, decreasing as n increases and $\dim(\Theta)$ decreases¹. The right side of (1) is the risk certificate, which guarantees an upper bound on the risk of generalization. All terms on the right side are also computable. Different bounding methods, for example VC and Pac-Bayes bounds, produce different complexity terms. In totality, certificates above 1 (with $l \in [0, 1]$) are

¹PAC-Bayes bounds compute expected risk over a posterior on θ and include a complexity term from divergence to a prior. If we confine the posterior to concentrate at a point and use a flat prior, the form resembles (Eq.1). Some bounds relate R and r nonlinearly, but can be approximated by (Eq.1) without affecting our argument.

vacuous, while we refer to those below 1 (or sometimes less than the random guess risk in the classification setting) as non-vacuous.

Traditional models like linear SVM achieve non-vacuous bounds through proper regularization (Vapnik, 1995; Shalev-Shwartz and Ben-David, 2014). However, conventional deep neural networks trained by gradient descent lack non-vacuous bounds unless n is extremely large, due to high $\dim(\Theta)$ from numerous *continuous* parameters. Even sparse adapter methods (e.g., LoRA (Hu et al., 2021), LoRA-XS (Bałazy et al., 2024)) face challenges from the continuous nature of Θ . No existing deep learning approach achieves meaningful generalization bounds in low-shot settings.

2.2. Meta-Learning

We consider *few-shot transfer learning*. In the upstream phase, we have training tasks $T_1, \dots, T_N$ sampled i.i.d. from an unknown distribution over family of tasks $p_{\text{true}}(T)$, each with data to train a task-specific PEFT adapter. In the downstream phase, we receive a new task $T^* \sim p_{\text{true}}(T)$ observed only through a small support set $S^* = \{(x_i, y_i)\}_{i=1}^n$ with n small (e.g., 8 or 16 examples). The goal is to adapt to T^* using only S^* . This follows standard meta-learning assumptions (Hospedales et al., 2021; Finn et al., 2017): downstream tasks are drawn from the same distribution as upstream tasks.

In standard meta-learning, the upstream phase learns meta-knowledge about task adaptation—such as a good initialization point (Finn et al., 2017), a metric space (Snell et al., 2017), or an optimization procedure—that enables rapid adaptation to new tasks using only the small support set S^* .

The challenge. Gradient-based adaptation on low-shot data S^* leads to vacuous bounds due to high hypothesis space complexity. We need a hypothesis space Θ that is both (i) rich enough to contain good solutions for new tasks, and (ii) structured enough to yield non-vacuous bounds with small n (Ravi and Larochelle, 2017).

3. Proposed Approach

3.1. High-Level Overview

Figure 1 contrasts standard meta-learning with our approach. Traditional methods (e.g., MAML (Finn et al., 2017)) learn meta-knowledge such as good initialization points, then adapt via gradient descent on downstream tasks. In contrast, STEEL learns the *manifold of task solutions*—capturing where good adapters live in parameter space—then adapts by sampling and selecting from this learned distribution.

This paradigm shift enables two key advantages: (i) **Gradient-free adaptation**, avoiding the optimization on

small downstream datasets that leads to overfitting, and (ii) **Finite hypothesis spaces**, making classical PAC-Bayes bounds applicable for the first time in few-shot deep learning.

3.2. Method Details

Upstream: Learning the manifold over task solutions. Given upstream task adapters $\bar{\Theta} = \{\theta_i\}_{i=1}^N$ where each θ_i solves task T_i , we treat these as samples from the true task distribution $p_{\text{true}}(T)$. Rather than meta-learning an adaptation procedure, we directly model the distribution of solutions via density estimation over $\{\theta_i\}_{i=1}^N$.

We construct the downstream hypothesis space Θ using one of two strategies:

- **Model zoo:** Set $\Theta = \bar{\Theta}$ directly. As N increases, $\bar{\Theta}$ better approximates $p_{\text{true}}(T)$, but requires storing all N adapters.
- **Diffusion model (primary):** Train a diffusion model (Ho et al., 2020) to learn $p(\theta)$ from $\bar{\Theta}$, then sample Θ from this model. This offers (i) *compactness*—one generative model versus N adapters, and (ii) *interpolation*—generating novel adapters between training tasks, often improving coverage of $p_{\text{true}}(T)$. Recent work (Wang et al., 2024; Du et al., 2023; Zhang et al., 2024) demonstrates diffusion models effectively capture structure in neural network weight spaces.

Downstream: Selection over finite hypotheses. Given test task support set $S^* = \{(x_i, y_i)\}_{i=1}^n$, few-shot adaptation is done by evaluate-then-select:

$$\theta^* = \arg \min_{\theta \in \Theta} r(\theta) = \frac{1}{n} \sum_{(x,y) \in S^*} l(\theta; x, y) \quad (2)$$

This is selection from a discrete set, not continuous optimization. The hypothesis space Θ encodes upstream inductive bias: each candidate is a plausible solution drawn from the learned task distribution.

However, the computational question naturally arises: *How do we solve (2) efficiently?* We consider two solutions:

- **Exhaustive search.** Evaluate $r(\theta)$ for all $\theta \in \Theta$ and select the one with the lowest loss. This guarantees finding the optimal θ^* , but is often computationally intractable (e.g., LLMs require a prohibitive number of forward passes or generations due to large $|\Theta|$).
- **Hierarchical search.** Use a tree-based approach such as hierarchical clustering of Θ . Evaluate losses at the top level (e.g., cluster centroids or medoids), select the best cluster, and recurse within it. This reduces computation to $O(\log |\Theta|)$ and can yield a good approxima-

tion to θ^* , but may result in suboptimal empirical loss $r(\theta)$ due to early pruning.

3.3. Generalization Guarantees

The finite hypothesis space enables tight PAC-Bayes bounds. With a uniform prior over Θ and Dirac delta posterior at θ^* , we obtain (Sec. 2.1.3 in (Alquier, 2021)) that with probability at least $1 - \epsilon$:

$$R(\theta) \leq r(\theta) + C \cdot \sqrt{\frac{\log \frac{|\Theta|}{\epsilon}}{2n}} \quad \text{for any } \theta \in \Theta \quad (3)$$

where $R(\theta) = \mathbb{E}_{(x,y) \sim T^*} [l(\theta; x, y)]$ is generalization risk and C bounds loss ($0 \leq l \leq C$). n is the total number of samples used and $|\Theta|$ is the total number of samples adapters from the diffusion model in our case.

The complexity term scales as $\sqrt{\log |\Theta|}$, so even $|\Theta| = 10^4$ maintains tight bounds. This makes our approach practical: a rich hypothesis space that represents the task distribution while yielding non-vacuous certificates. The bound is minimized by choosing θ^* with lowest empirical risk, directly justifying Eq. 2.

Computational considerations. For large $|\Theta|$, we use either *exhaustive search* (evaluate all candidates; optimal but expensive) or *hierarchical search* (cluster-based with $O(\log |\Theta|)$ complexity; faster but may select suboptimal θ). The bound (Eq. 3) holds regardless; hierarchical search may yield higher $r(\theta)$, slightly loosening but never invalidating the certificate. See Appendix D for pseudo-code.

3.4. Training Assumptions and Bounds Validity

As in standard meta-learning, we assume upstream and downstream tasks are sampled i.i.d. from the same underlying task distribution. This ensures that the hypothesis space learned during the upstream phase contains high-quality solutions for unseen downstream tasks, a common assumption in few-shot transfer learning. We validate this empirically over a wide set of benchmarks demonstrating comparative downstream performance to SGD.

Importantly, the PAC-Bayes bound is valid without assuming downstream tasks match the upstream distribution. This assumption only affects tightness: distribution shift likely increases empirical risk, which loosens but doesn't invalidate the bound. Empirically, we find this is not an issue under mild distribution shifts. Crucially, even under distribution shift, the bound provides a principled criterion for deployment decisions—we can evaluate whether the certified risk is acceptable before deploying. This form of risk assessment is not available in gradient-based meta-learning, where adaptation always produces a solution without quantifying its reliability.

The bound’s validity is also independent of the selection procedure used to choose θ^* from Θ . Eq. 3 holds for any $\theta \in \Theta$; exhaustive search, hierarchical search, or any heuristic (e.g., the two-stage procedure used for LaMP, Appendix B.2) only affects *which* adapter is returned and thus $r(\theta^*)$. A suboptimal selection loosens the certificate but never invalidates it, and the certificate is always computed on the finally deployed adapter.

4. Related Work

Risk certificates. While traditionally applied to simple models, recent work extends risk certificates to deep learning. Notable approaches include using data-dependent priors (Perez-Ortiz et al., 2021) and parameter quantization of PEFT adapters (Lotfi et al., 2024) in tabula-rasa and single source transfer learning respectively. Nevertheless, these still require large training sets to obtain non-vacuous bounds. In the meta-learning literature, Zakerinia et al. (2024) proposed meta-learning PAC-Bayes bounds, which aims to facilitate tighter generalisation bounds by extracting a common prior from up-stream tasks in a multi-task setting. However, it was only demonstrated on toy problems and doesn’t scale to large models due to the memory requirements of nested gradient computations.

Model diffusion methods. Several works explore diffusion for generating model parameters: NNDiffusion (Wang et al., 2024) for BN modules, ProtoDiff (Du et al., 2023) for ProtoNet-based few-shot learning, and MetaDiff (Zhang et al., 2024) and D2NWG (Soro et al., 2025) for gradient-free meta-learning. Scaling properties of Diffusion based learning were also explored (Schürholt et al., 2024). However, none provide risk certificates for the generated models.

Sparse adapter (PEFT) methods. Sparse adapters reduce learnable parameters, crucial for our diffusion-based sampling. While LoRA (Hu et al., 2021) uses trainable low-rank matrices and VeRA (Kopiczko et al., 2024) uses fixed matrices with trainable diagonals, we adopt LoRA-XS (Bałazy et al., 2024), which uses SVD with a trainable full matrix for the singular values. Such adapters have facilitated large-data guarantees (Lotfi et al., 2024), but in our framework they will facilitate low-shot guarantees.

Connection to our Approach. In terms of assumptions, we consider the same multi-task setting of meta-learning (Hospedales et al., 2021), model-zoo (Huang et al., 2024) and model diffusion (Wang et al., 2024) approaches. They all aim to facilitate downstream learning by knowledge transfer from upstream tasks. In terms of solution we share the benefit of model-zoo and model-diffusion methods in being able to use third-party off-the-shelf pre-trained upstream models, rather than requiring an expensive joint learning procedure like meta-learners. Like meta-learners

and model-diffusers, but unlike zoo methods, we do learn a task-agnostic component (the diffusion model).

5. Experiments

Metrics for Certified Learning Quality: In the low-shot learning context, we run many learning episodes with different random small training sets (Wang et al., 2020). Thus we need evaluation metrics for the typical empirical performance and certificate strength instead of a single accuracy and certificate for one large scale learning (Lotfi et al., 2024). This is straightforward for empirical test performance: we report the average over episodes of the relevant task metric (e.g., accuracy, RMSE). For certificates, we report the minimum (best), median (typical) and maximum (worst) case error guarantee over all downstream tasks. Since many learners often produce vacuous certificates in the low-shot regime (e.g., guaranteed error not below 1), we use *the proportion of tasks which have a non-vacuous certificate* as our leading metric. Finally, to quantify how tight the certificates are, we also report *gap* - the average distance between certificates and query errors over all faithful bounds. More concretely, this is computed as $\sum_{i=0}^K \frac{\max(0, \text{bound}_i - \text{query error}_i)}{K}$ where K is the total number of downstream tasks.

Competitors: For single task methods that exploit only the foundation model and the target downstream task, we compare standard SGD, along with MeZO (Malladi et al., 2023) and BBPT (Yu et al., 2023) – two state of the art gradient-free learners – for fine-tuning. For multi-task alternatives that exploit also the same task distribution as STEEL, we compare the model zoo learner LoRA-Hub (Huang et al., 2024) and the meta-learner MetaPB (Zakerinia et al., 2024). LoRA-Hub randomly samples from the model zoo and learns a new adapter as a linear combination of sampled adapters. Note that all methods are already combined with the same (experiment-specific) PEFT strategy for fair comparison. For SGD, MeZO, BBPT and MetaPB, this is an upgrade provided by us - they would definitely otherwise fail in the full parameter space. For SGD, MeZO, BBPT and LoRA-Hub, we combine them with recent quantization bound of (Lotfi et al., 2024) (Eq. 7) to provide guarantees, MetaPB uses their own PAC Bayes bound, and STEEL uses the bound in Eq. 3.

5.1. Guarantees for few-shot LLM adaptation

Datasets: We use the LaMP personalization benchmark (Salemi et al., 2024) for low-shot LLM adaptation. LaMP consists of fixed training (seen) and evaluation (unseen) clients, each with support and query data. We evaluate on three datasets: LaMP-2 (nominal classification: movie tagging), LaMP-3 (ordinal classification: product rating), and LaMP-5 (text generation: scholarly title generation).

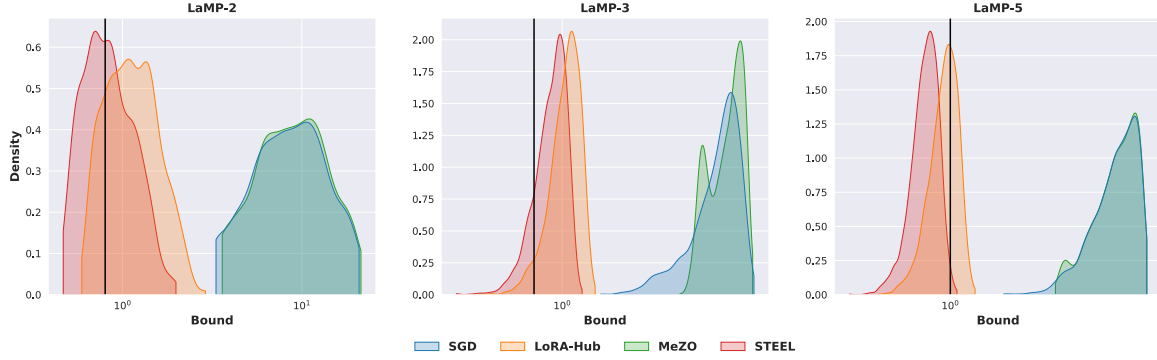


Figure 3. Distribution of generalisation guarantees (x-axis, log scale) obtained over few-shot LLM adaptation episodes. Vertical lines indicate the vacuous bound threshold. STEEL provides a dramatically better distribution of provable generalisation outcomes compared to alternatives.

Table 1. LaMP LLM adaptation benchmark results.

		SGD	LoRA-Hub	MeZO	STEEL
LaMP-2	% Non-Vacuous Tasks	0.00%	32.13%	0.00%	65.12%
	Average Gap	9.03	0.38	9.27	0.36
	Min Bound	3.32	0.59	3.60	0.47
	Median Bound	8.52	1.12	8.85	0.80
	Max Bound	20.86	2.90	21.36	1.99
	Accuracy↑	63.25%	57.51%	63.30%	63.74%
	F1↑	56.15%	50.84%	57.03%	55.69%
LaMP-3	% Non-Vacuous Tasks	0.00	5.00	0.00	15.48
	Average Gap	2.98	0.25	3.06	0.26
	Min Bound	1.35	0.51	2.54	0.43
	Median Bound	3.56	1.04	3.76	0.93
	Max Bound	4.75	1.30	4.53	1.17
	MAE↓	0.217	0.230	0.242	0.231
	RMSE↓	0.511	0.526	0.531	0.524
	Cross-Entropy↓	0.479	0.739	0.626	0.693
LaMP-5	% Non-Vacuous Tasks	0.00	63.84	0.00	99.16
	Average Gap	3.95	0.38	3.96	0.28
	Min Bound	1.61	0.52	2.58	0.40
	Median Bound	4.57	0.96	4.57	0.81
	Max Bound	5.86	1.25	5.88	1.06
	ROUGE-1↑	47.04%	47.05%	47.03%	47.22%
	ROUGE-L↑	42.79%	42.75%	42.73%	42.89%

Setup: Following prior work (Tan et al., 2024; Salemi et al., 2024), we fine-tune Flan-T5 base (Chung et al., 2024) on all seen users’ support data to build a task-specific base model. Then, we personalize it by training a LoRA-XS module per user (Bałazy et al., 2024), with rank 6 and alpha 16 (2592 tunable parameters). These modules, trained on the same support data used for base fine-tuning, form our model zoo. For efficiency, STEEL uses hierarchical search (Section 2). Hyperparameters are in Appendix B.2.

Results: Our main contribution relates to the ability to provably certify the generalisation of low-shot learning. In terms of low-shot LLM adaptation, Figure 3 visualises the distribution of certification outcomes over a large number of episodes for the three LAMP benchmarks. Taking note of the log-scale on the x-axis for generalisation guarantee strength, we can see that our STEEL learner provides dramatically better guarantees than conventional continuous-

parameter learner alternatives, thanks to its discrete hypothesis space. The vertical lines indicate the threshold for vacuous bounds. Standard learners such as SGD and MeZO have no mass left of the threshold, while a substantial number of STEEL learning episodes are non-vacuously guaranteed. Table 1 provides more detailed quantitative results in terms of various metrics. Notably, to assess provable generalisation, the data visualised in Figure 3 is summarised as the % non-vacuous metric (the fraction of episodes which have guarantees with a strength above chance-level), and the median guarantee strength across episodes.

From Table 1 results we can see that: (1) Standard supervised learning approaches such as (gradient-based) SGD and (gradient-free) MeZO have no non-vacuous episodes - no few-shot learning task can be guaranteed. (2) STEEL has the most non-vacuous episodes for each benchmark, with almost every few-shot learning episode being guaranteed in the LAMP-5 benchmarks. The median STEEL episode also has a substantially non-vacuous guarantee for all three benchmarks. (3) Interestingly, LoraHUB combined with Lotfi et al. (2024)’s discretization bound also has some non-vacuous episodes, but less than STEEL. (3) STEEL has comparable or better empirical test accuracy compared to existing approaches such as SGD and MeZO, while providing a huge improvement in certifiability.

5.2. Guarantees for few-shot visual recognition

Datasets: We use fine-grained vision datasets with standard seen/unseen class splits: CUBirds (Wah et al., 2011), FGVC-Aircraft (Maji et al., 2013), Describable Textures (Cimpoi et al., 2014), and Flowers-101 (Nilsback and Zisserman, 2008). Splits follow Meta-Dataset (Triantafillou et al., 2020) for Flowers and Learn2Learn (Arnold et al., 2020) for the others.

Setup: We sample k -shot, 5-way tasks following meta-learning conventions (Triantafillou et al., 2020), evaluating at $k \in \{1, 2, 4, 8, 16\}$. CLIP (Radford et al., 2021) serves

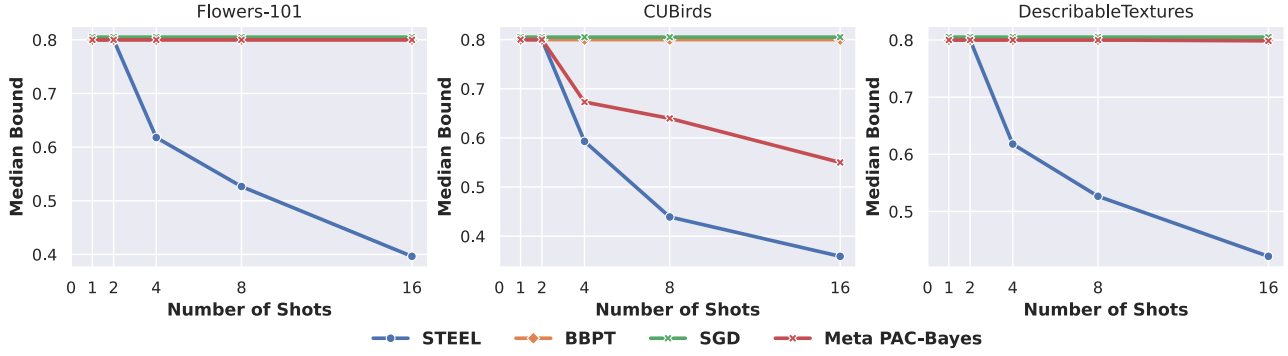


Figure 4. Dependence of generalisation guarantee on training set size. Our finite-hypothesis class learner STEEL achieves non-vacuous guarantees from 4-shot onward. Standard approaches provide no guarantees anywhere in this low-shot range.

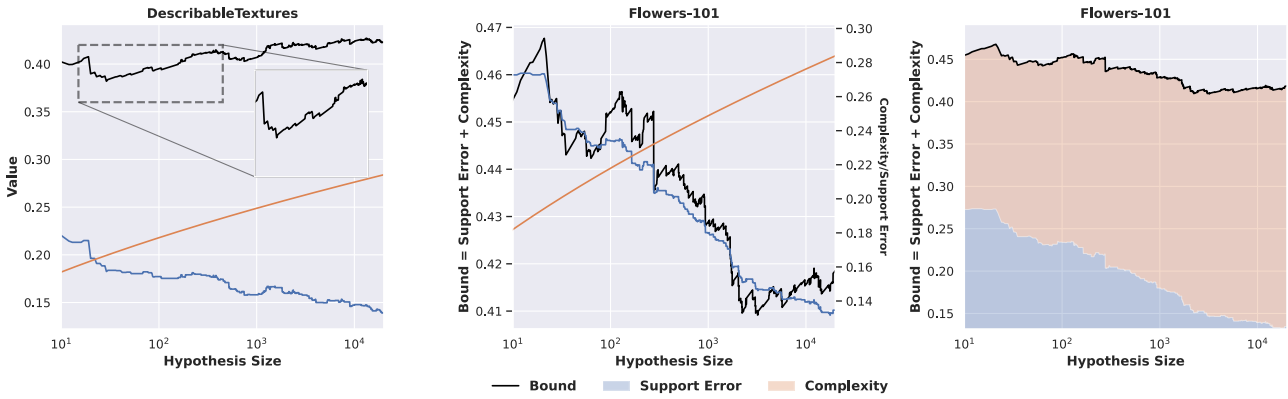


Figure 5. “Learning curves” illustrating empirical and certified learning dynamics of STEEL with respect to samples/iterations, which is equivalent to hypothesis space size. More samples improves the training (support) error, while increasing the complexity penalty. The sum of these two terms instantiates the generalisation guarantee (Eq. 3) achieved for a given number of samples.

as the backbone. For each task, we sample 5 classes, draw n support examples per class, and adapt using CoOp (Zhou et al., 2022)—i.e., prompt tuning with a 2-token prefix per class (1024 total tunable parameters). The model zoo is built from training classes; evaluation uses disjoint unseen classes. Due to fast inference, we perform exhaustive search over sampled prompts. Meta-training, evaluation, and bound details for Meta-PB are provided in Appendix A.3. Hyperparameters for all other vision experiments are in Appendix B.3.

Results: The results in terms of mean training and testing error versus complexity are summarised for three datasets in Figure 2. The dots for each learner reflect the training set sizes of 1, 2, 4, 8, and 16-examples per-class, and the white/grey zone separation delineates the space of non-vacuous vs vacuous bound outcomes. The main message is, among all methods evaluated, our finite hypothesis class approaches consistently achieve the strongest non-vacuous guarantees across this range of training set sizes. Every result for the standard SGD and BBPT approaches is vacuous and cannot be guaranteed. Please note, Meta-PB is not included as it is not directly comparable on the plot

y-axis (empirical error term); Meta-PB uses the unbounded cross-entropy loss (Equation 20) as the risk metric, while our and other competing methods use empirical error.

For the 16-shot case, these experiments are quantified in Table 2. Similarly to the results for LLM adaptation, we can see the dramatic difference in % of non-vacuous episodes, and dramatic improvement in the min, median and max bound obtained over episodes. Compared to the LLM case, STEEL pays a slightly higher price in terms of empirical test accuracy compared to SGD for some benchmarks, however this is small compared to the stark difference in certification performance. The state of the art Meta-PB, also oriented at low-shot certification in some cases (CUB) manages to non-vacuously certify most episodes. However, in all cases the strength of this certification is much worse than STEEL (gap, median bound).

Figure 4 highlights the evolution of the median generalisation bound as a function of the training set size. For STEEL it becomes non-vacuous from 4-shots onward, and the standard approaches never become non-vacuous²

²Note their bound is substantially worse than 0.8, but for simple

Table 2. Aggregate over 16-Shots 5-way visual recognition episodes. Standard Errors are reported in Appendix C.4

Method	SGD	BBPT	Lora-Hub	Meta-PB	STEEL
CUBirds					
Non-Vacuous Ratio	0.00%	0.00%	100.00%	97.50%	100.00%
Average Gap	2.49	2.48	0.41	0.45	0.24
Min Bound	2.55	2.55	0.44	0.34	0.30
Median Bound	2.58	2.59	0.52	0.55	0.36
Max Bound	2.64	2.68	0.59	0.85	0.47
Average Accuracy	90.32%	89.27%	89.45%	89.24%	88.40%
Describable Textures					
Non-Vacuous Ratio	0.00%	0.00%	100.00%	50.00%	100.00%
Average Gap	2.43	2.46	0.42	0.54	0.24
Min Bound	2.55	2.55	0.54	0.57	0.36
Median Bound	2.55	2.63	0.65	0.80	0.42
Max Bound	2.58	2.78	0.74	1.06	0.53
Average Accuracy	87.95%	83.20%	77.40%	81.12%	81.50%
FGVCAircrafts					
Non-Vacuous Ratio	0.00%	0.00%	35.00%	0.00%	97.50%
Average Gap	2.31	2.45	0.41	0.86	0.22
Min Bound	2.58	2.64	0.63	0.87	0.45
Median Bound	2.64	2.80	0.83	1.25	0.61
Max Bound	2.78	3.01	1.05	1.76	0.85
Average Accuracy	65.57%	62.37%	57.85%	59.84%	61.37%
Flowers-101					
Non-Vacuous Ratio	0.00%	0.00%	87.50%	10.00%	100.00%
Average Gap	2.51	2.50	0.42	0.68	0.27
Min Bound	2.55	2.55	0.45	0.65	0.31
Median Bound	2.55	2.59	0.62	1.15	0.40
Max Bound	2.55	2.70	1.04	1.77	0.61
Average Accuracy	95.90%	90.15%	78.10%	71.23%	84.90%

5.3. Further Analysis

Learning Curves: We discuss and provide some insight into the learning process of our discrete hypothesis class learner. Standard gradient-descent takes repeated update steps to find a model that better fits a training set. By analogy, our gradient-free STEEL draws more samples as it attempts to iteratively sample a model that better fits the training set. Our main experiments use a fixed number of 20,000 samples on all vision datasets. Figure 5 illustrates our learner’s behaviour by showing the equivalent of a learning curve for our model. The x-axis is the number of samples drawn, and equivalently the learning theoretic hypothesis space size. Unlike SGD, this means that there is a direct dependence of hypothesis class complexity ($|\Theta|$ in Eq. 3) and the number of iterations/samples. This is reflected in the steadily increasing red complexity curve in Figure 5(left, middle). We can also see that the training/support error goes down consistently over iterations/samples as the sampler progressively discovers better models. The generalisation bound (black line) is given by the sum of the training error and complexity. The figure illustrates one case (Flowers, middle, right) where the bound continues to improve up to a large number of samples/hypothesis size, because the continued improvement in training error outweighs the complexity gain. It also illustrates a case (DTD, left) where the training error improvement is slower and quite rapidly outweighed by the complexity gain, so that the best bound is actually achieved after quite a small number of samples.

Sampler vs Zoo: STEEL compresses the upstream pre-trained models into a learned model generator. Selecting from the upstream models using downstream task performance visualisation, we plot it as chance-level for 5-way classification.

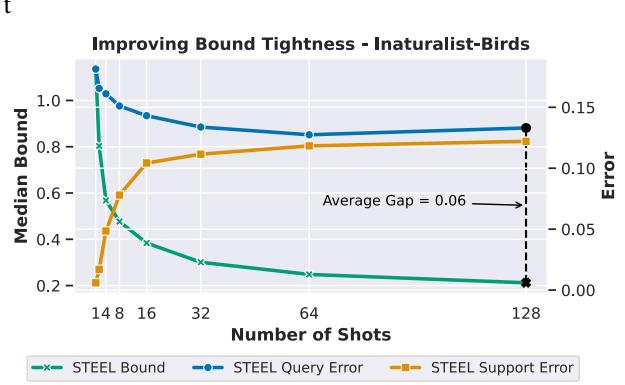


Figure 6. Support/query error and certified risk vs. support set size on iNaturalist birds. As the number of shots increases, support and query errors converge, while the bound continues to tighten. The gap between the certified risk and empirical query error drops to 6% at 128 shots, demonstrating the ability to produce tight certificates.

mance as a criterion provides an alternative approach to learning that also corresponds to a finite hypothesis space. Our generator approach was motivated by ensuring scalability with respect to a large number of upstream models, and also to improve accuracy by enabling interpolation between upstream models rather than solely being limited to selecting one of them. Figure 2 shows that STEEL’s diffusion sampler tends to provide improved accuracy compared to its raw model zoo, especially for Flowers and DTD. On average across all datasets STEEL consistently outperforms Model Zoo. Detailed per-dataset per-shot performance is deferred to Appendix C, Table 5. Furthermore, we show in Appendix C.1 how to certify zero-shot CLIP using confidence intervals, and demonstrate that STEEL not only achieves tighter guarantees but also substantially outperforms it in terms of empirical performance.

Tightening STEEL Certificates: So far, we have focused on the very low-shot regime to highlight our key result: the first non-vacuous certificate in this challenging setting. However, many applications may require stronger guarantees. To explore this, we increase the support set size from very low to moderately low shot using the iNaturalist dataset (Horn et al., 2018) (birds subset, excluding overlapping classes), with a diffusion model pre-trained on CUB. This setup allows scaling up to 128-shot evaluation. Figure 6 shows: (1) As the training set grows, the support error increases initially as there are more data points to fit, but soon stabilizes. The finite hypothesis-space does not problematically under-fit an increasingly large training set. (2) Query error decreases and saturates around 64 shots. (3) Crucially, the STEEL bound keeps tightening: as support size grows, the complexity term continues to shrink even after support error plateaus, resulting in a certified risk just 6% above empirical test error at 64–128 shots—while using far fewer examples than standard certification approaches (Lotfi et al., 2024).

(4) Finally, it is important to note that this strong result is *despite* the upstream-downstream distribution shift (CUB $\rightarrow$ iNaturalist), showing that STEEL is robust to some task distribution shift.

STEEL Out-of-Distribution Robustness Appendix F provides a controlled study under graded in-domain corruption (CUB-C) and a head-to-head comparison against MAML, further substantiating STEEL’s robustness under distribution shift.

Diffusion Ablation: Appendix E presents ablation experiments comparing our diffusion model to alternative generative approaches. Results show that diffusion outperforms these baselines, supporting its use in our method.

6. Discussion and Conclusion

STEEL introduces a paradigm replacing continuous adaptation with discrete hypothesis selection: it trains a diffusion model on upstream task adapters to learn the “solution manifold,” then adapts to new tasks by sampling and selecting the best adapter based on support set loss. This shift from continuous to finite hypothesis spaces enables tight PAC-Bayes bounds, achieving the first non-vacuous generalization guarantees for few-shot learning with large models while maintaining comparable empirical accuracy. Across LLM personalisation and fine-grained vision benchmarks, STEEL delivers the highest fraction of non-vacuous certificates in the low-shot regime, often by a wide margin, while gradient-based, zeroth-order, and meta-PAC-Bayes baselines remain mostly vacuous at $n \leq 16$ shots. This establishes discrete hypothesis selection as a viable paradigm for certified foundation-model adaptation—not a theoretical curiosity, but a practical method.

The remainder of this section sketches the research programme this paradigm opens—tightening the certificate, closing the accuracy gap, and extending to cross-domain transfer—with full development in Appendix G.

Tightening the certificate. Two directions promise tighter bounds without algorithmic change. The $\log |\Theta|$ term in Eq. 3 treats every sampled adapter as contributing independently to complexity, but many produce near-identical predictions; replacing $|\Theta|$ with the prediction-space covering number $\mathcal{N}(\Theta, \varepsilon)$, computed via clustering on upstream reference data, decouples certificate strength from raw sample count. Separately, the diffusion model’s own likelihood provides a natural data-dependent prior: substituting $-\log p_{\text{diff}}(\theta^*)$ for $\log |\Theta|$ tightens the bound whenever θ^* lies in a high-likelihood region—precisely when upstream and downstream distributions are aligned. Both repurpose information STEEL already computes (Appendix G.1).

Closing the accuracy gap. STEEL’s certificate validity

depends only on sampled adapters’ empirical risk, not on the generator architecture or PEFT family. This makes the framework a passive beneficiary of progress in adjacent fields: stronger weight-space architectures such as SANE (Schürholt et al., 2024) and improved PEFT modules translate directly to lower $r(\theta^*)$ and tighter bounds. Most distinctively, the discrete-hypothesis framework supports *mixture hypothesis classes*: training separate diffusion models over different PEFT families (LoRA-XS, VeRA, prompt tuning) and constructing Θ as their union enables implicit per-task PEFT selection at only $\log K$ additional complexity cost. A complementary lever is *sample-then-refine*: select θ^* via STEEL, then apply bounded continuous updates within an ε -ball, yielding a hybrid discrete–continuous procedure for which PAC-Bayes bounds with data-dependent priors remain applicable (Wang et al., 2019).

Cross-domain transfer. The cross-domain few-shot literature has converged on the view that robustness to domain shift is driven by the breadth of meta-training rather than by test-time adaptation (Li et al., 2021; Liu et al., 2021; Eustratiadis et al., 2024). STEEL is structurally well-suited to this regime: per-task adapters are trained independently before aggregation, so scaling to additional source domains amounts to extending the upstream pool—no episode balancing, no nested optimisation. Support-set-conditional generation pushes further by steering sampling toward downstream-relevant regions of weight space—the generative analogue of gradient-based adaptation, within a finite hypothesis class; the same mechanism also tightens empirical fit in-domain (Appendix G.3).

A new paradigm and what it opens. Gradient-based meta-learning evolved incrementally over nearly a decade: from MAML (Finn et al., 2017) on single-domain benchmarks (2017), to Meta-Dataset (Triantafillou et al., 2020) (2020), to methods designed for cross-domain transfer (Li et al., 2021; Liu et al., 2021). STEEL introduces a fundamentally different paradigm and establishes its feasibility together with the first non-vacuous certificates for few-shot deep learning. The directions above are the open frontier this paradigm unlocks. Establishing scaling laws for weight-space generative models—analogue to those now well-understood for language models—would let practitioners predict the certificate–accuracy frontier as a function of upstream compute, hypothesis-space size, and downstream support-set size, and is one of the more exciting longer-horizon questions opened by this work.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, and we mention a non-exhaustive concerns here. First, for LLM experiments, please note that the diffusion model is trained on adapters trained on user-specific data. We can use the diffusion model to interpolate between adapters to potentially avoid distributing the private model zoo among clients, but this does have potential for data leakage by memorization (Staab et al., 2024). It is worth investigating privacy preserving methods for such concerns (Miranda et al., 2024).

On the vision side of our experiments, despite our benchmark being on fine-grained classification tasks of publicly available datasets, we necessitate the reminder of ethical concerns in computer vision as our method is easily applicable across domains and applications (Waelen, 2023).

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A. Existing Risk Bounds for Deep Models

A.1. Vanilla PAC-Bayes Bound

This is the vanilla, non-transfer learning bound. As a baseline, one can also contrast with the vanilla PAC-Bayes bound (i.e., non-meta learning bound). This essentially follows the Cantoni’s bound, and can be written as follows. With probability at least $1 - \epsilon$, the following holds:

$$\mathbb{E}_{Q(\theta)}[R(\theta)] \leq \mathbb{E}_{Q(\theta)}[r(\theta)] + \sqrt{\frac{\text{KL}(Q||\pi) + \log(1/\epsilon)}{2n}} \quad (4)$$

Here n is the test support size. We can set $\pi = \mathcal{N}(0, \kappa^2 I)$ for some fixed κ_π and $Q(\theta) = \mathcal{N}(\mu, \Sigma^2)$ where the parameters (μ, Σ) can be learned by minimizing the right hand side. The sampled version $\theta^z = \mu + \sqrt{\Sigma} \cdot z$, $z \sim \mathcal{N}(0, I)$ can be used during the optimization. Once optimized, the minimum value of the right hand side serves as the error bound for the test task.

A.2. Bound with Parameter-Level Quantization

In (Lotfi et al., 2024), they proposed a non-vacuous bound for the LLM based on the model parameter quantization (e.g., fixed-length floating point machine representation). There are several differences to our approach:

1. The paper is about LLM pre-training setup with large training data, and the bound would be vacuous if training data size is not large enough (e.g., $\geq 10K$).
2. They derive the same finite hypothesis space PAC-Bayes bound, but replace the $\log |H|$ term by $\log(1/p(h))$ where $p(h)$ is the prior likelihood, and $\log(1/p(h))$ is approximated and upper-bounded by $C(h)$ which is the number of bits for representing the hypothesis h .
3. The finite hypothesis space comes from the fixed-size floating point representation for real numbers (e.g., if there are d trainable parameters, then $C(h) = d \cdot 32$), but to reduce it further, they propose what is called the SubLoRA, which is a random subspace representation (i.e., $\theta = Pw$, P = random subspace basis, w = coefficients) of the LoRA A/B matrices.
4. Also, instead of 32 bit for each of d params, they do some clustering to reduce it to shorter coding, more precisely the arithmetic coding.

The followings are some details of their bound derivation. With probability at least $1 - \epsilon$,

$$R(\theta) \leq r(\theta) + C \cdot \sqrt{\frac{K(\theta) + 2 \log K(\theta) + \log(1/\epsilon)}{2n}} \quad (5)$$

where $K(\theta)$ is the Kolmogorov complexity bound that can be estimated as:

$$K(\theta) = \sum_{i=1}^d (\# \text{ of bits in the arithmetic coding of } \theta_i) \quad (6)$$

where $d = \dim(\theta)$ for the PEFT parameters θ . The arithmetic coding requires clustering of parameters θ_i s, thus being dependent on the particular θ used. However, we can consider the best (i.e., the tightest) bound possible. That is, even if we have 1 bit for every θ_i (the minimal code length possible), $K(\theta) = d$, and plugging this into (5) yields:

$$R(\theta) \leq r(\theta) + C \cdot \sqrt{\frac{d + 2 \log d + \log(1/\epsilon)}{2n}} \quad (7)$$

which is the *best* scenario. Please note that this should be interpreted in the context of currently known bounds rather than all possible bounds.

In (5), as before, $R(\theta) = \mathbb{E}_{z \sim T^*}[l(\theta; z)]$ is the generalization error of θ , $r(\theta) = \frac{1}{n} \sum_{z \in S^*} l(\theta; z)$ is the empirical error on the support data with size $n = |S^*|$, and C is the maximal loss value (i.e., $0 \leq l \leq C$).

A.3. PAC-Bayes Meta Learning Bound

In (Zakerinia et al., 2024) they proposed an extension of the PAC-Bayes bound for meta learning. Their meta learning algorithm aims to learn a distribution over the adaptation algorithms $A(S) \rightarrow Q(\theta)$ where S is the support training data for a task and $Q(\theta)$ is the posterior distribution in the PAC-Bayes theorem. In their paper, any adaptation algorithm A is allowable provided that A internally makes use of a prior distributions in the PAC-Bayes theorem (e.g., the adaptation algorithms that try to minimize the PAC-Bayes bound), so that $A(S)$ becomes a function of the prior. And the meta learning model is a distribution ρ that can sample the prior, and the meta learning amounts to learning the distribution ρ .

Although they provided a concrete example in Appendix B.2 in their paper, the described meta training involves an optimization of a loss function that contains an optimal value of another problem, which is typically attained by gradient descent. Just like MAML-type algorithms with a large number of inner gradient steps, this renders the application of their theorem to large-scale model/data impractical. So we propose a more practical reptile-like adaptation strategy, essentially admitting a closed-form inner optimal solution, that is practical in large model/data scenarios. Note that under our setting the meta learning PAC-Bayes bound is still applicable since they said the theorem works for *any* adaptation algorithms.

. CONCRETE CASES WITH (REPTILE-LIKE) QUADRATIC-REGULARIZED ADAPTATION ALGORITHMS

Let θ be the parameters of a neural network or PEFT parameters. Denote $\dim(\theta)$ by d . The adaptation algorithm $A(S)$ returns a PAC-Bayes posterior $Q(\theta) = \mathcal{N}(\mu, \Sigma)$ for some task support data S as input. We consider diagonal Σ . The algorithm A itself is parametrized by the PAC-Bayes prior $P_1(\theta) = \mathcal{N}(\mu_1, \Sigma_1)$ since we only deal with those A s that internally use P_1 as a part of the algorithm without dependency on any other things. One candidate for A is the algorithm that minimizes the PAC-Bayes bound with prior P_1 with respect to $Q(\theta)$. However, as said earlier, this incurs MAML-like optimization that is impractical.

Our choice of A is as follows: (Step-1) We first find $(\mu^{SGD}, \Sigma^{SGD}) = \arg \min_{\mathcal{N}(\theta; \mu, \Sigma)} \mathbb{E}[l(\theta; S)]$ by gradient descent. Technically, this involves evaluating $l(\theta; S)$ with reparametrized $\theta = \mu + \sqrt{\Sigma}z$ for $z \sim \mathcal{N}(0, I)$, where the gradients take: $\nabla_{\mu} l(\theta; S) = \nabla_{\theta} l(\theta; S)$ and $\nabla_{\Sigma} l(\theta; S) = \nabla_{\theta} l(\theta; S) \cdot 0.5 \Sigma^{-0.5} z$ by the chain rule. So it is as tractable as SGD since we only need conventional backprop $\nabla_{\theta} l(\theta; S)$ to get gradients with respect to (μ, Σ) . (Step-2) Solve the following quadratic optimization:

$$(\mu^*, \Sigma^*) = \arg \min_{\mu, \Sigma} \|(\mu, \Sigma) - (\mu^{SGD}, \Sigma^{SGD})\|^2 + \alpha \cdot \|(\mu, \Sigma) - (\mu_1, \Sigma_1)\|^2 \quad (8)$$

The intuition is that the adapted model (μ, Σ) has to be close to the fitted $(\mu^{SGD}, \Sigma^{SGD})$ and also close to the prior (μ_1, Σ_1) . The trade-off coefficient α is user's choice, but when inspired by the PAC-Bayes bound objective, we can set $\alpha = 1/\sqrt{|S|}$. The main benefit of this quadratic-regularized adaptation algorithm is that we have the closed-form solution:

$$\mu^*(\mu_1, S) = \frac{1}{1 + \alpha} \mu^{SGD} + \frac{\alpha}{1 + \alpha} \mu_1 \quad (9)$$

$$\Sigma^*(\Sigma_1, S) = \frac{1}{1 + \alpha} \Sigma^{SGD} + \frac{\alpha}{1 + \alpha} \Sigma_1 \quad (10)$$

Note that (μ^*, Σ^*) are functions of the prior (μ_1, Σ_1) , and so we used the function notation in (9–10) together with dependency on S although μ^* does not depend on Σ_1 and Σ^* similarly. We treat $(\mu^{SGD}, \Sigma^{SGD})$ as constant.

Next we define the meta prior (π) and posterior (ρ) distributions for the meta learning PAC-Bayes bound. As suggested in (Zakerinia et al., 2024),

$$\pi(A) = \mathcal{N}((\mu_1, \Sigma_1); 0, \kappa_{\pi}^2 I_{2d}) \quad (11)$$

$$\rho(A) = \mathcal{N}((\mu_1, \Sigma_1); M_1, \kappa_{\rho}^2 I_{2d}) \quad (12)$$

where $M_1 = (M_1^{\mu}, M_1^{\Sigma})$ is the parameters for the meta posterior ρ . Here κ_{π} and κ_{ρ} are the fixed scales, and in their paper they used $\kappa_{\pi} = 100$ and $\kappa_{\rho} = 10^{-3}$. Finally, they assume delta meta prior and posterior in the complexity term: $\mathbb{P}(A) = \mathbb{Q}(A) = \delta_{P_1}$.

This leads to the following meta training objective:

$$\begin{aligned} \min_{M_1} \hat{R}(\rho) + & \underbrace{\sqrt{\frac{\text{KL}(\rho||\pi) + \log(4\sqrt{N}/\epsilon)}{2N}}}_{=:T_1} + \\ & \underbrace{\sqrt{\frac{\text{KL}(\rho||\pi) + N\hat{E}(\rho) + \log(8nN/\epsilon) + 1}{2nN}}}_{=:T_2} \end{aligned} \quad (13)$$

where

$$\begin{aligned} \hat{R}(\rho) &= \mathbb{E}_{A \sim \rho} \left[\frac{1}{N} \sum_{i=1}^N l(A(S_i); S_i) \right] \\ \hat{E}(\rho) &= \mathbb{E}_{A \sim \rho} \left[\frac{1}{N} \sum_{i=1}^N \text{KL}(A(S_i) || P_1) \right] \\ \text{KL}(\rho || \pi) &= \frac{2d\kappa_\rho^2 + \|M_1\|^2}{2\kappa_\pi^2} - d + 2d \log \frac{\kappa_\pi}{\kappa_\rho} \end{aligned}$$

Recall that n is the support data size and N is the number of meta training tasks.

Meta training procedure (stochastic approximation version). Since it is nearly infeasible to compute the objective for all N tasks (for the purpose of one gradient update for M_1), we adopt the stochastic approximate optimization. That is, for each sampled task batch B (out of N tasks), we compute the followings (stochastic estimates of the objective and the gradient):

- $\hat{T}_0 \approx \hat{R}(\rho)$ where

$$\hat{T}_0 = \frac{1}{|B|} \sum_{i \in B} l(\theta^{z_i}; S_i), \quad (14)$$

$$\theta^{z_i} = \mu^*(\mu_1, S_i) + \sqrt{\Sigma^*(\Sigma_1, S_i)} \cdot z_i, \quad z_i \sim \mathcal{N}(0, I_d) \quad (15)$$

$$\mu_1 = M_1^\mu + \kappa_\rho \cdot z^\mu, \quad z^\mu \sim \mathcal{N}(0, I_d) \quad (16)$$

$$\Sigma_1 = M_1^\Sigma + \kappa_\rho \cdot z^\Sigma, \quad z^\Sigma \sim \mathcal{N}(0, I_d) \quad (17)$$

Here $\mu^*(\mu_1, S_i)$ and $\Sigma^*(\mu_1, S_i)$ are determined by (9–10). Note that $\hat{T}_0$ is a function of M_1 , and the gradient of $\hat{T}_0$ with respect to M_1 can be obtained from $\nabla_{\theta^z} \hat{T}_0$ by the chain rule similarly as described previously.

- T_1 as it is since it is easy closed-form.
- $\hat{T}_2 \approx T_2$ where $\hat{T}_2$ only replaces $\hat{E}(\rho)$ by the following from T_2 :

$$\hat{E}(\rho) \rightarrow \frac{1}{|B|} \sum_{i \in B} \text{KL}(A(S_i) || P_1) \quad (18)$$

$$= \frac{1}{|B|} \sum_{i \in B} \text{KL}(\mathcal{N}(\mu^*, \Sigma^*) || \mathcal{N}(\mu_1, \Sigma_1)) \quad (19)$$

This has a closed form from the Gaussian KL formula.

As the above steps allow us to compute the meta training objective and its gradient, we can update M_1 , and move on to a next batch.

Test-time bound. Once we have meta-learned M_1 , we can compute the error bound for a new task that comes with the support data S^* . In essence, we solve the following optimization problem:

$$Q^* = \arg \min_Q \hat{R}(Q) + \sqrt{\frac{\text{KL}(Q||P_1) + \log(8n/\epsilon) + 1}{2n}} \quad (20)$$

We parametrize $Q = \mathcal{N}(\mu, \Sigma)$ and solve it with respect to (μ, Σ) . First, $\hat{R}(Q)$ can be approximated by $l(\theta^z; S^*)$ with $\theta^z = \mu + \sqrt{\Sigma} \cdot z$, $z \sim \mathcal{N}(0, I_d)$. We can use the chain rule similarly for $\nabla_{\mu, \Sigma} \hat{R}(Q)$. In the KL term, P_1 has to be sampled from ρ (i.e., our learned M_1). We can use $P_1 = \mathcal{N}(M_1^\mu + \kappa_\rho \cdot z^\mu, M_1^\Sigma + \kappa_\rho \cdot z^\Sigma)$ and compute the KL using the Gaussian KL formula. Once optimized, the minimum value of (20) becomes the error bound for this test task which holds with probability at least $1 - \epsilon$.

B. Architecture and Training Recipes

B.1. Diffusion Architecture and Training Recipe

First, the diffusion model forward encoder uses a 1000 timesteps with a linear scheduler over noise between $1e-4$ to $2e-2$. For the decoder, we use an MLP network for the diffusion model with 3 hidden-layers. The hidden layer dimension is $4 \times$ the size of its input. This is 10,240 for LaMP (divisible by 512 for parallelization concerns) and 4096 for vision. A layer conditioned time embedding of the diffusion step is added (summed with) the hidden layer’s hidden representation.

The time embedding is generated from the diffusion timestep using sinusoidal embeddings as per the original DDPM model (Ho et al., 2020). The dimensionality of the sinusoidal embedding is equal to the Diffusion MLP hidden dimension. Subsequently, the embedding is transformed using a two-layer MLP with first layer expanding the dimension to $4 \times$ the network’s hidden dimension and the second layer downscaling again to original hidden dimension. For example, on the vision experiments, the time embedding network has hidden dimension of 4098×4 . Finally, to condition the time embedding computed by the two-layer MLP time embedding network for each layer, we apply a different linear transformation per diffusion hidden layer.

We train the diffusion model for 30K epochs for all experiments with a batch size of 1,024. We use the LAMB optimizer (You et al., 2020) with a learning rate of 0.01. For vision experiments, we found that we can continue improving performance if we continue training for a second stage of 10K more epochs. For the second stage, we use a one-cycle learning rate scheduler (Smith and Topin, 2018) with default hyperparameters (as found in pytorch). The maximum learning rate starts from 0.0004 reaching a maximum of 0.001 over a 1000 steps. We keep an exponential-moving average of the network weights throughout training with a decaying rate of 0.9999.

B.2. Flan-T5 + LaMP Hyperparameters

For training the base model across all datasets, we use LaMP’s original recipe (Salemi et al., 2024). We use a batch size of 64, AdamW optimizer with a learning rate of $5e-5$ and weight decay of size 0.0001. We use a linear warmup for the learning rate over 5% of the total number of training steps.

To build the model zoo, we found that we required to tune Adam optimizer learning rate and per-dataset epochs per-dataset. We optimizer the hyperparameters to improve performance on the training split (seen users) query data. We use learning rates of 0.01, 0.01, 0.0001 and 20, 10, 10 epochs for LaMP-2, LaMP-3 and LaMP-5 respectively. For all datasets, we used a linear warmup for the learning rate over 5% of the total number of training steps. We used the same recipe to train a per-user LoRA-XS model on the unseen users/novel tasks.

For LoRA-Hub, we used default hyperparameters as proposed by the original authors. First, we sample 20 random adapters from the model zoo. The weights of the linear combination is initialized with zeros and truncate min/max weights to -1.5/1.5. We do maximum inference steps of 40 with NeverGrad default hyperparameters.

Finally, we transform the SGD number of epochs to MeZO. The authors used $32 \times$ as many epochs as SGD in the original paper (Malladi et al., 2023). This translates to 640, 320, 320 epochs for LaMP-2, LaMP-3, and LaMP-5. We tested the three learning rates proposed to search over by the authors. We fixed a learning rate of $1e-3$ across all datasets because we consistently found $1e-4$ to not learn and $1e-2$ to be unstable.

For LaMP, we sample 10K LoRA-XS modules from the diffusion model. We use k-means clustering on the diffusion samples to produce N clusters where N is chosen as the minimum Silhouette score. We evaluate N between [2,150] inclusive.

For each cluster, we find the medoid; the adapter closest to the centroid of the cluster. During evaluation, we choose a cluster and evaluate all adapters therein. From the cluster, we short-list the best 15 adapters using the Flan-T5 training loss. On the best 15 adapters, we use text generation to produce an answer with greedy sampling. Using the LaMP benchmark proposed model selection metric for each dataset, we select the “winning” adapter.

The two-stage selection (training-loss short-list, then benchmark metric on generations) is a practical heuristic for efficiency. As discussed in Section 3.4, the certificate is computed on the finally deployed adapter, and the bound in Eq. 3 remains valid regardless of the selection rule; a suboptimal selection only loosens $r(\theta^*)$ and thus the certificate, but never invalidates it.

. BOUND METRICS

For support errors, we use 1-Accuracy for LaMP-2, and ROUGE-1 for LaMP-5. For LaMP-3, both RMSE and MAE are not bounded. Therefore, we devise a cross-entropy like metric for the dataset. First, we convert the ordinal vectors to one-hot encodings. Subsequently, we calculate the absolute error between the labels one-hot encoding and Flan-T5 model logits, and divide by 2. This guarantees the error to be bounded in the [0-1] range inclusive. We use this metric as support error term.

LaMP-2 Intricacies: LaMP benchmark has one query sample per user. For LaMP-3 and LaMP-5, this suffices since the generated support error is continuous. Nevertheless, for LaMP-2, the accuracy term, which we use to derive the support error, becomes the 0/1 loss. Therefore, we split the support data in novel tasks to support and query data with ratio 80% and 20% respectively. If the split generates only 1 query samples, we move one sample from support to query to have a minimum of two-samples in query. We use the same split for all evaluations across SGD, MeZO, LoRA-Hub and our proposed methods. For reproducibility, all splits were done deterministically. Furthermore, we did not constrain the split to have same classes across both support and query. LaMP classification tasks are long-tailed. Therefore, for a novel task, a user might have classes X and Y in support but the query ends up with classes A and B making it a more challenging benchmark for all methods.

Finally, we truncate the support sizes of LaMP-3 and LaMP-5 to 256 samples across all methods. This is done deterministically for reproducibility. The reason for truncating the dataset is pure computational concerns.

. MODEL ZOO SIZE

For LaMP, we build a model zoo by training one PEFT adapter per-task in the dataset. Each user is treated as one task. This yields 3820, 20,000, and 9,682 total adapters/tasks in LaMP-2, LaMP-3, and LaMP-5 respectively.

. COMPUTE RESOURCES

All experiments were conducted on 10xA40 GPUs. The base model finetuning for generating task specific models took a couple of hours with a batch size of 64 using 2 GPUs. Time to train an adapter for one user using one GPUs varies from dataset to dataset and from user to user depending on their available dataset size. This takes as low as 10 seconds and up to 2 mins per user. Diffusion model training on the collected model zoo varied among datasets too since each epoch was of different total number of steps because of different dataset sizes. We used one A40 GPU per diffusion model and train using the recipe described earlier.

Meanwhile, downstream adaptation using STEEL varies between datasets. For LaMP-2, this averaged 0.8 minutes per user (50 A40 GPU hours total across all users). For comparison, SGD adaptation takes 10 seconds to 2 minutes per user. Similarly, MeZo requires approximately 2.5x more wall-clock time than SGD (it needs 7x more epochs to converge while being 3x faster per epoch), making STEEL faster than MeZo for language tasks. LaMP-5 averaged 7.6 minutes per user (315 A40 GPU hours total) since it requires text generation painting it slower.

B.3. CLIP + CoOP Hyperparameters

To build the model zoo, we used the authors original hyperparameters to train CoOP because we found them to work the best. This is SGD with a learning rate of 0.002. For Flowers-101, we train for 200 epochs. For DTD, FGVC Aircraft and CUBirds, we trained for up to 300 steps and found a One Cycle learning rate useful to stabilize training. These same hyperparameters were used to evaluate SGD on novel tasks. For BBPT, we use the authors default hyperparameters (Yu et al., 2023). We found that the method converges within 8000 “API call”. We attempted to run for a budget of 20K as our diffusion model offers but found that performance did not improve. Please note that the original authors reduce the dimensionality of the prompt using a small network because evolutionary optimization struggles in high-dimension. They reduce dimensionality

to 512. Nevertheless, since we train only 2-tokens (dimensionality=1,024), then we do not use the small dimensionality reduction network.

Finally, for MetaPB, we train the upstream model for 20 epochs on our model zoo which takes 24 hours similarly to our diffusion model training. We use Adam with default learning rate of 0.01 for both upstream and downstream. Downstream adaptation was done for 300 epochs with same Adam optimizer setting. We train both means and variances. Variances are clipped to 0.1 for numerical stability.

For vision experiments, we found that exhaustive search was fast enough even though we sample 20K adapters from the diffusion model.

Bound Metrics: we use $1 - \text{Accuracy}$ as the support error term in our bound calculation for all vision experiments.

Model Zoo Size: We randomly sample 16-shot 5-way tasks for building the model zoo from each respective dataset. We train 10,000 total tasks per-dataset for the model zoo. The diffusion model is trained on this model zoo. Once trained, we sample the diffusion model once and fix the samples across all downstream evaluation for 1, 2, 4, 8 and 16 shots.

Compute Resources: For SGD, training one task upstream for model zoo collection or downstream for gradient descent adaptation takes 10-15 seconds. BBPT downstream takes 10-15 seconds as well. Training the diffusion model on a model zoo, since we fix model zoo size in CLIP experiments, was the same among all experiments. Each diffusion model took 24 A40 GPU hours to train. STEEL takes 30 second to 1 minute downstream for adapting to a new task. Finally, meta-training phase of MetaPB takes 24 hours similarly to our diffusion model training. MetaPB downstream takes 15 second per task too.

C. Extra Vision Results

C.1. Certifying Zero-Shot Performance

Table 3. Comparing median bounds between STEEL and zero-shot CLIP given a 16-shots support set.

Median Bound	Flowers	CUB	DTD	FGVCA
STEEL	0.40 (+0.03)	0.36 (-0.06)	0.42 (+0.11)	0.61 (+0.10)
ZS CLIP	0.43	0.30	0.53	0.71

One might assume that zero-shot CLIP is a strong *certified* baseline because it is evaluated without any fine-tuning, and the support and query sets are drawn i.i.d. from the same distribution. This can lead to the impression that its support-set accuracy reflects test-time performance. However, these are still empirical estimates and do not account for uncertainty due to sampling. In contrast, our certificates explicitly bound the generalization gap from limited support data. To facilitate a fairer comparison, we compute a Langford-style test-set bound (Theorem 3.3 by [Langford \(2005\)](#)) using zero-shot CLIP’s support accuracy from Table 3, treating it as a proxy for true performance. The resulting guarantees are included in Table C.1. Although zero-shot CLIP performs well empirically, our STEEL method achieves tighter guarantees and often stronger accuracy in the few-shot setting.

C.2. Effect of Candidate Pool Size

We conduct an ablation to test whether STEEL’s gains come from the diffusion model’s learned distribution or simply from evaluating more candidate adapters in vision tasks. To investigate this, we compare three conditions: selecting from the 10K model zoo directly, generating and evaluating 10K samples from STEEL, and generating and evaluating 20K samples from STEEL. Results on 16-shot vision tasks are shown in Table 4.

C.3. Accuracy Results Across Shots

Results are provided in table 5.

C.4. Vision Results with Standard Errors (STE)

We report the performance of the 16-shot 5-way vision experiments in Table 6. For the non-vacuous ratio, average gap, and median bounds, we compute the standard error using the bootstrap method. Specifically, we sample with replacement and

Dataset	Zoo (10K)	STEEL (10K)	STEEL (20K)
CUBirds	87.97	<u>88.30</u>	88.40
DescribableTextures	79.25	<u>81.42</u>	81.50
FGVCAircrafts	61.02	61.50	<u>61.37</u>
Flowers-102	82.92	<u>84.22</u>	84.90
Average	77.79	<u>78.86</u>	79.04

Table 4. Comparison of candidate pool size and source. STEEL at 10K samples already outperforms selecting from the 10K zoo, indicating gains come from the learned distribution rather than pool size. Increasing to 20K provides marginal additional improvement.

report the standard deviation of the resulting distribution as the estimated standard error. For accuracy, computing STE is straightforward.

D. STEEL Pseudo-Code

We provide pseudo-code in Algorithm 1. There are upstream and downstream learning phases, as per meta-learning, model-zoo, and other transfer learning workflows. We refer to these as meta-training and meta-testing, to borrow terminology from the meta-learning literature (Hospedales et al., 2021; Wang et al., 2020). The upstream meta-training phase trains one model per-task and then treats these models as data to learn a weight-space diffusion model. This effectively compresses upstream models into a single model generator that provides a prior finite hypothesis class. The downstream meta-testing phase performs low-shot learning by sampling from the generator and returning the model that minimises the downstream empirical risk. This leads strong generalisation guarantees in practice, because the finite hypothesis class incurs a low complexity cost in the certificate calculation.

Algorithm 1 Transfer Learning via Diffusion and Finite Hypothesis Space

Meta-Training Phase:

- 1: **Input:** Training tasks $\{T_1, T_2, \dots, T_N\}$ sampled i.i.d. from $p_{\text{true}}(T)$
- 2: **Output:** Trained diffusion model $p(\theta)$
- 3: **for** $i = 1$ to N **do**
- 4: Train PEFT adapter on task T_i to obtain optimal parameters θ_i
- 5: $\bar{\Theta} \leftarrow \bar{\Theta} \cup \{\theta_i\}$ {Collect task-specific parameters}
- 6: **end for**
- 7: Train diffusion model $p(\theta)$ using parameter set $\bar{\Theta} = \{\theta_1, \theta_2, \dots, \theta_N\}$ as training data

Meta-Testing Phase:

- 1: **Input:** New task $T^* \sim p_{\text{true}}(T)$ with support set $S^* = \{(x_i, y_i)\}_{i=1}^n$
 - 2: **Input:** Trained diffusion model $p(\theta)$, hypothesis space size $|\Theta|$
 - 3: **Output:** Selected parameters θ^* and generalization bound
 - 4: Sample finite hypothesis space: $\Theta = \{\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(|\Theta|)}\} \sim p(\theta)$
 - 5: **Evaluate-then-Select Strategy:**
 {Can use hierarchical search}
 - 6: **for** $j = 1$ to $|\Theta|$ **do**
 - 7: Compute empirical risk: $r(\theta^{(j)}) = \frac{1}{n} \sum_{(x,y) \in S^*} l(\theta^{(j)}; x, y)$
 - 8: **end for**
 - 9: Select optimal parameters: $\theta^* = \arg \min_{\theta \in \Theta} r(\theta)$
 - 10: **Compute Risk Certificate:**
 - 11: With probability $\geq 1 - \epsilon$, the generalization error satisfies:
 - 12: $R(\theta^*) \leq r(\theta^*) + C \cdot \sqrt{\frac{\log \frac{|\Theta|}{\epsilon}}{2n}}$
-

E. Diffusion Model Ablation

The use of a diffusion model in STEEL is motivated by recent work on generative modeling in weight space (Wang et al., 2024), which demonstrates that diffusion models can effectively capture the complex, multimodal distribution of neural network parameters. To evaluate whether this modeling capacity is critical for performance, we conduct an ablation study comparing against two alternative generative baselines for sampling parameter candidates in the LoRA-XS subspace:

- **Random Sampling:** Each parameter is sampled independently from a Gaussian distribution whose mean and standard deviation are estimated from the corresponding statistics of the model zoo. This baseline assumes a fully factorized, unimodal distribution and requires no training or hyperparameter tuning.
- **Gaussian Mixture Model (GMM):** A GMM with diagonal covariance matrices is trained on the model zoo parameters. The number of mixture components is selected from $\{2, 4, 8, 16, 32, 64\}$ using the Bayesian Information Criterion (BIC). Once trained, the GMM is used to sample candidate parameter vectors.

The results in Table 7 demonstrate that the diffusion model significantly outperforms both random sampling and GMM-based sampling. Random sampling performs poorly, with zero-accuracy. Generated data from this baseline are random tokens. GMM improves over random but still fails to outperform the zero-shot baseline, likely due to its limited capacity to model high-dimensional parameter correlations and multimodality. In contrast, the diffusion model exceeds the performance of SGD-based fine-tuning, despite requiring no training on the test-time user data. This highlights its strong generative prior over parameter space.

F. Robustness Under Distribution Shift

A natural question for any few-shot transfer learner is how it behaves when downstream tasks deviate from the upstream task distribution. Section 3.4 established that the STEEL certificate (Eq. 3) remains *valid* under distribution shift – the i.i.d. assumption affects only *tightness*, not correctness – and Figure 6 demonstrated this on the CUB→iNaturalist cross-domain transfer. Here we provide a more controlled study: (i) STEEL under graded, in-domain corruption (CUB-C), and (ii) a head-to-head comparison against MAML, the canonical gradient-based meta-learner whose inner-loop adaptation is often presumed to confer OOD robustness.

F.1. Setup: CUB-C Corruption Benchmark

Inspired by ImageNet-C (Hendrycks and Dietterich, 2019), we construct **CUB-C** by applying three perturbations – Gaussian noise, Gaussian blur, and brightness reduction – to CUB images at three severity levels (mild, moderate, severe). This yields downstream tasks drawn from a distribution progressively further from the clean upstream tasks while remaining within the same semantic domain (fine-grained bird classification). All other meta-test settings match main text (16-shot, 5-way; CLIP backbone; CoOp prompt tuning; exhaustive search over $|\Theta| = 20\text{K}$ STEEL samples). The diffusion model is the same one trained on clean CUB used throughout the main paper; no re-training is performed.

F.2. STEEL Under Graded Distribution Shift

Table 8 reports STEEL’s certificate behavior across severities. Three observations are salient.

(1) Certificate validity is preserved. STEEL maintains a 100% non-vacuous ratio across all severity levels, including severe corruption that degrades empirical accuracy by ~ 11.7 percentage points ($88.60\% \rightarrow 76.95\%$). This empirically confirms the theoretical claim of Section 3.4: distribution shift inflates empirical risk $r(\theta)$, which loosens the right-hand side of Eq. 3, but does not invalidate the certificate.

(2) The bound degrades gracefully. The median bound rises from 0.35 (clean) to 0.45 (severe). The minimum bound is essentially flat ($0.28 \rightarrow 0.30$), reflecting the complexity term – which depends only on $|\Theta|$ and the support size – being unaffected by the shift. The maximum bound widens more substantially ($0.43 \rightarrow 0.80$), tracking the heaviest-tailed empirical-risk increases.

(3) Certificates remain a useful deployment signal. A practitioner inspecting a median bound of 0.45 on severely corrupted data receives a principled, quantitative input to the deploy/no-deploy decision – a signal that is structurally unavailable from

uncertified methods, since empirical accuracy on a small support set is itself an unreliable estimator of test performance in the few-shot regime (Shimabucoro et al., 2024).

F.3. Head-to-Head: STEEL vs. MAML Under Shift

A common intuition is that gradient-based meta-learners such as MAML (Finn et al., 2017) should outperform discrete-selection methods like STEEL under distribution shift, since the inner-loop SGD step in principle allows adaptation to move away from the meta-learned initialization. We test this directly by comparing the two methods on CUB-C under matched conditions (identical CLIP backbone, CoOp adapter, 16-shot 5-way episodes, same severity levels). MAML certificates are computed using the quantization bound of Lotfi et al. (2024) (Eq. 5), which is the tightest known bound applicable to its continuous hypothesis space.

Two findings stand out (Table 9).

MAML provides no certificate at any severity. Its median bound remains in the 2.6–2.7 range across all conditions, well above the vacuous threshold. The continuous hypothesis space of MAML’s adapted weights incurs a complexity term that small- n certificates cannot overcome, irrespective of the empirical accuracy attained.

MAML’s empirical accuracy advantage is negligible and non-monotonic in shift severity. The gap is ~ 1 pp at mild shift, effectively zero at moderate shift, and *reverses* at severe shift (MAML 76.47% vs. STEEL 76.95%). This directly contradicts the hypothesis that inner-loop SGD confers substantial OOD adaptation. MAML’s adaptation is constrained to a local neighborhood of the meta-learned initialization – whether by a small number of inner steps or by an explicit regularizer (e.g., iMAML (Rajeswaran et al., 2019)) – and without such locality, adaptation reduces to plain SGD on a very small support set, which generalizes poorly. Neither approach *fully* addresses OOD; STEEL’s discrete selection delivers comparable accuracy while preserving certifiability.

F.4. Scope and Limitations

We deliberately scope this study to in-domain corruption rather than cross-dataset transfer (e.g., miniImageNet→CUB). The CUB-C setup isolates the effect of input-distribution shift while holding the semantic task distribution fixed, which is the cleanest setting for assessing the *validity* and *graceful degradation* of the certificate. Cross-dataset OOD additionally entails task-distribution shift and is known to be challenging for the entire transfer/meta-learning paradigm; it also requires retraining the upstream diffusion model on a different source, which is outside the scope of our current study. We discuss this honestly as a limitation and outline concrete extensions of STEEL to this setting in Section G, including: (i) conditioning the diffusion model on support-set embeddings to steer generation toward downstream-relevant regions while preserving the finite hypothesis space; (ii) hybrid hypothesis classes that union diffusion models trained on multiple source domains; and (iii) combining STEEL body selection with a lightweight gradient-based classifier head, at small additional certificate cost.

F.5. Summary

The certificate guarantee underlying STEEL is robust to in-domain distribution shift in the operational sense that matters for deployment: certificates remain valid, degrade gracefully with empirical risk, and provide a quantitative signal that uncertified methods structurally cannot. Against MAML – the prototypical gradient-based meta-learner – STEEL matches or exceeds empirical accuracy under moderate-to-severe shift while providing the only non-vacuous certificate of the two methods. Together with the CUB→iNaturalist result of Figure 6, these experiments substantiate the claim that STEEL’s certifiability is not purely an in-distribution phenomenon.

G. Limitations and Future Directions

STEEL establishes the feasibility of certifiable few-shot transfer learning with large models. Like any new paradigm, it leaves open a research program for tightening guarantees, closing the empirical-accuracy gap with continuous-hypothesis methods, and extending coverage beyond the in-distribution setting. We discuss these directions in three groups: tightening the certificate (Sec. G.1), improving empirical accuracy (Sec. G.2), and extending STEEL to cross-domain and out-of-distribution (OOD) settings (Sec. G.3). All directions preserve the structural property that gives STEEL its guarantee – a tractable hypothesis space – and are therefore drop-in modifications rather than rethinks of the framework.

G.1. Tightening the Certificate

Covering-number refinement of the hypothesis space. The bound in Eq. 3 treats every sampled adapter as contributing independently to the complexity term $\log |\Theta|$. In practice, many sampled adapters produce near-identical predictions and should not both count toward complexity. What matters for generalisation is not $|\Theta|$ but the *covering number* of Θ in prediction space: the minimum number of ε -balls needed to cover the induced predictor set under a suitable functional distance. A clean and immediate route to exploiting this within our existing bound is to pre-process Θ into an ε -net $\Theta' \subseteq \Theta$, obtained by clustering predictions on a held-out upstream reference set and retaining one representative per cluster. Running STEEL on Θ' leaves Eq. 3 unchanged but replaces $\log |\Theta|$ with $\log |\Theta'| = \log \mathcal{N}(\Theta, \varepsilon)$. Because the reference set is upstream data, no dependence on the downstream support set S^* is introduced, preserving the assumptions of the bound. Practitioners can then draw *more* diffusion samples (improving empirical fit) while the effective hypothesis-space size grows only with prediction-space diversity rather than raw sample count. Pushing this further would require switching bound technology – e.g., chaining or fat-shattering arguments – which is a more ambitious extension.

Data-dependent priors over Θ . Our certificate uses a uniform prior over the sampled set Θ , which yields the $\log |\Theta|$ complexity term in Eq. 3. A tighter alternative is to use the diffusion model’s own likelihood as a non-uniform prior: $\pi(\theta) \propto p_{\text{diff}}(\theta)$ for $\theta \in \Theta$. The PAC-Bayes complexity term then becomes $\text{KL}(\delta_{\theta^*} \parallel \pi) = -\log \pi(\theta^*)$, which replaces $\log |\Theta|$ with $-\log p_{\text{diff}}(\theta^*) + \log Z$ (with $Z = \sum_{\theta \in \Theta} p_{\text{diff}}(\theta)$). Whenever the selected θ^* lies in a high-likelihood region of the diffusion model – which is precisely what we expect when the upstream and downstream task distributions are aligned – this term is smaller than $\log |\Theta|$ and the bound tightens. Because the diffusion model is trained on a disjoint upstream split, π is independent of the downstream support set S^* , preserving the assumptions of the bound (cf. Perez-Ortiz et al. (2021)). A practical consideration is that exact diffusion log-densities are intractable; however, the standard DDPM ELBO is a *lower* bound on $\log p_{\text{diff}}(\theta^*)$, equivalently an *upper* bound on $-\log p_{\text{diff}}(\theta^*)$, and substituting it into the certificate yields a valid (if slightly conservative) bound. The normalising constant Z can likewise be upper-bounded using upstream samples. This direction repurposes information that STEEL already computes – the diffusion likelihood – to tighten the certificate without changing the algorithm.

G.2. Improving Empirical Accuracy

Stronger weight-space architectures. STEEL’s certificate validity depends only on the empirical risk of sampled adapters, not on the architecture of the generator. Recent weight-space learning methods such as SANE (Schürholt et al., 2024) model neural-network parameter distributions substantially better than the MLP-based diffusion model used in our experiments. Substituting such architectures into the upstream phase is a drop-in improvement to sample quality – and therefore to downstream accuracy – without any change to the theoretical framework.

Better PEFT modules. Our experiments fix the PEFT method per backbone (LoRA-XS (Bałazy et al., 2024) for LLMs; CoOp (Zhou et al., 2022) for vision). The choice of PEFT method affects both individual adapter quality and the structure of the parameter distribution the diffusion model must learn. Unified PEFT benchmarks (Xin et al., 2024; Mai et al., 2025) document significant method-dependent variation in accuracy. Since lower adapter quality translates directly to higher $r(\theta)$ and looser bounds, swapping in stronger PEFT families as the landscape evolves is a practical lever for improving both accuracy and certificate tightness simultaneously.

Mixture hypothesis classes across PEFT families. Currently, a single diffusion model is trained over a single PEFT family. A natural extension is to train *separate* diffusion models over different PEFT families and construct the downstream hypothesis space as their union. The complexity term increases only by a small constant ($\log K$ for K families), while the hypothesis space gains substantially richer coverage. At meta-test time, STEEL automatically selects whichever PEFT family best fits the support set, providing implicit task-conditional PEFT selection at no asymptotic cost in certificate strength.

G.3. Cross-Domain and OOD Extensions

Appendix F shows that STEEL’s certificate behaves well under graded in-domain corruption (CUB-C). Cross-*dataset* transfer – where upstream and downstream tasks differ in both input statistics and label semantics – is more challenging for the entire transfer/meta-learning paradigm. We discuss three concrete directions for extending STEEL to this regime, each of which preserves certifiability.

Scaling meta-training across diverse domains. The cross-domain few-shot learning literature has converged on a consistent

finding: robustness to domain shift is driven primarily by the breadth of the meta-training distribution rather than by the choice of meta-test adaptation procedure. Methods such as URL (Li et al., 2021) and tri-M (Liu et al., 2021) achieve strong cross-domain performance using *frozen* or near-frozen test-time adaptation, succeeding because their upstream phase aggregates over many domains. Recent comparisons of single- vs. multi-source meta-dataset training (Eustratiadis et al., 2024) confirm this picture. STEEL is structurally well-suited to multi-domain upstream training: because per-task adapters are trained independently before being aggregated into the diffusion model, scaling to additional domains amounts to extending the upstream pool. There is no need for the careful multi-domain episode balancing required by gradient-based meta-learners. The diffusion model can then learn a multi-modal distribution with modes per source dataset, and at meta-test it automatically selects samples in the vicinity of the most-related source mode.

Support-set-conditional generation. STEEL currently samples from the unconditional prior $p(\theta)$. Conditioning the diffusion model on a representation of the support set S^* – e.g., a permutation-invariant encoding of $\{(x_i, y_i)\}_{i=1}^n$ – would steer generation toward downstream-relevant regions of weight space. This is the generative-model analogue of MAML’s gradient-based adaptation: it uses the support set to inform *where* to search, but does so within a finite hypothesis space sampled from $p(\theta \mid S^*)$. Certifiability is preserved (the hypothesis space remains finite), with a modest bound loosening to account for the conditioning channel that can be analysed via standard PAC-Bayes conditioning arguments.

Sample-then-refine: hybrid discrete–continuous adaptation. A natural way to combine STEEL’s certifiability with the local adaptability of gradient-based methods is to select θ^* from Θ as in STEEL, and then apply bounded SGD updates within an ε -ball around θ^* . The resulting hypothesis space is a finite union of bounded continuous regions, for which PAC-Bayes bounds with data-dependent priors remain applicable. A particularly tractable instantiation is the SimpleShot-style decomposition (Wang et al., 2019): select the backbone/body via STEEL (preserving the discrete certificate over body parameters), then fit a linear classifier head with SGD on the support set. The classifier head admits a tight separate bound – linear classifiers in low-shot regimes are exactly the setting where classical generalisation theory is sharp – and the two bounds can be combined.

G.4. A Research Programme

We close by situating STEEL in the broader trajectory of meta-learning research. Gradient-based meta-learning evolved incrementally over nearly a decade: MAML (Finn et al., 2017) (2017) was demonstrated on single-domain benchmarks; Meta-Dataset (Triantafillou et al., 2020) (2020) introduced cross-domain evaluation; methods specifically designed for cross-domain transfer (URL, tri-M, TSA) followed in 2021–2022; and scaling laws for meta-learning have only recently come into view. STEEL introduces a fundamentally different paradigm – discrete hypothesis selection from a learned weight-space distribution – and establishes its feasibility together with the first non-vacuous certificates for few-shot deep learning. The directions above constitute a concrete and incremental research programme for extending this foundation: tighter certificates via prediction-space covering numbers and data-dependent priors; closing the empirical-accuracy gap via stronger weight-space architectures, better PEFT families, and mixture hypothesis classes; and reaching the cross-domain regime via multi-domain upstream scaling, conditional generation, and hybrid discrete–continuous adaptation. Establishing scaling laws for weight-space generative models – analogous to those now well-understood for language models – is a final, longer-horizon item that would let practitioners predict the certificate–accuracy frontier as a function of upstream compute, hypothesis-space size, and downstream support-set size.

Table 5. CLIP+CoOp few-shot learning. Accuracies over different number of shots.

Dataset	Zero Shot	SGD	BBPT	Lora-Hub	Meta-PB	Model Zoo	STEEL
1-Shots							
CUBirds	83.82%	81.77%	85.80%	86.45%	88.25%	86.00%	86.72%
DescribableTextures	67.29%	69.97%	74.10%	69.75%	78.55%	72.12%	73.97%
FGVCAircraft	47.44%	53.00%	55.05%	54.05%	56.46%	54.37%	55.40%
Flowers-101	81.14%	83.55%	80.40%	73.02%	54.20%	74.25%	76.75%
Avg	69.92%	72.07%	73.84%	70.82%	69.37%	71.69%	73.21%
2-Shots							
CUBirds	83.82	85.82%	86.92%	86.12%	88.40%	86.77%	86.50%
DescribableTextures	67.29	75.95%	76.57%	71.12%	78.99%	75.22%	76.17%
FGVCAircrafts	47.44	52.30%	57.90%	55.15%	58.31%	55.82%	57.30%
Flowers-101	81.14	86.92%	85.05%	73.82%	61.25%	77.17%	79.50%
Avg	69.92	75.25%	76.61%	71.56%	71.74%	73.75%	74.87%
4-Shots							
CUBirds	83.82	88.30%	88.42%	86.72%	88.77%	87.50%	87.47%
DescribableTextures	67.29	81.02%	77.90%	73.82%	78.85%	77.40%	79.82%
FGVCAircrafts	47.44	58.65%	60.55%	55.37%	58.26%	57.55%	60.10%
Flowers-101	81.14	91.95%	87.07%	77.50%	65.44%	79.82%	81.42%
Avg	69.92%	79.98%	78.49%	73.36%	72.83%	75.57%	77.21%
8-Shots							
CUBirds	83.82	89.75%	88.40%	87.22%	88.68%	87.42%	87.32%
DescribableTextures	67.29	85.07%	81.47%	75.82%	79.96%	79.90%	81.77%
FGVCAircrafts	47.44	62.07%	61.55%	57.40%	58.56%	58.77%	61.72%
Flowers-101	81.14	94.30%	88.67%	79.47%	68.72%	80.32%	82.52%
Avg	69.92%	82.80%	80.02%	74.98%	73.98%	76.61%	78.34%
16-Shots							
CUBirds	83.82	90.32%	89.27%	89.45%	89.24%	87.97%	88.40%
DescribableTextures	67.29	87.95%	83.20%	77.40%	81.12%	79.25%	81.50%
FGVCAircrafts	47.44	65.57%	62.37%	57.85%	59.84%	61.02%	61.37%
Flowers-101	81.14	95.90%	90.15%	78.10%	71.23%	82.92%	84.90%
Avg	69.92%	84.94%	81.25%	75.70%	75.36%	77.79%	79.04%

Table 6. Aggregate over 16-Shots 5-way visual recognition episodes with Standard Errors

Method	SGD	BBPT	Meta-PB	STEEL
CUBirds				
Non-Vacuous Ratio	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	97.50 $\pm$ 2.47	100.00 $\pm$ 0.00
Average Gap	2.49 $\pm$ 0.01	2.48 $\pm$ 0.01	0.45 $\pm$ 0.01	0.24 $\pm$ 0.01
Min Bound	2.55	2.55	0.34	0.30
Median Bound	2.58 $\pm$ 0.01	2.59 $\pm$ 0.01	0.55 $\pm$ 0.01	0.36 $\pm$ 0.01
Max Bound	2.64	2.68	0.85	0.47
Average Accuracy	90.32 $\pm$ 0.78	89.27 $\pm$ 0.50	89.24 $\pm$ 0.70	88.40 $\pm$ 0.72
Describable Textures				
Non-Vacuous Ratio	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	50.00 $\pm$ 7.91	100.00 $\pm$ 0.00
Average Gap	2.43 $\pm$ 0.01	2.46 $\pm$ 0.01	0.54 $\pm$ 0.01	0.24 $\pm$ 0.01
Min Bound	2.55	2.55	0.57	0.36
Median Bound	2.55 $\pm$ 0.00	2.63 $\pm$ 0.01	0.80 $\pm$ 0.02	0.42 $\pm$ 0.01
Max Bound	2.58	2.78	1.06	0.53
Average Accuracy	87.95 $\pm$ 0.58	83.20 $\pm$ 0.76	81.12 $\pm$ 0.68	81.50 $\pm$ 0.96
FGVCAircrafts				
Non-Vacuous Ratio	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	97.50 $\pm$ 2.47
Average Gap	2.31 $\pm$ 0.01	2.45 $\pm$ 0.01	0.86 $\pm$ 0.02	0.22 $\pm$ 0.01
Min Bound	2.58	2.64	0.87	0.45
Median Bound	2.64 $\pm$ 0.01	2.80 $\pm$ 0.02	1.25 $\pm$ 0.03	0.61 $\pm$ 0.02
Max Bound	2.78	3.01	1.76	0.85
Average Accuracy	65.57 $\pm$ 1.45	62.37 $\pm$ 1.60	59.84 $\pm$ 1.21	61.37 $\pm$ 1.25
Flowers-101				
Non-Vacuous Ratio	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	10.00 $\pm$ 4.74	100.00 $\pm$ 0.00
Average Gap	2.51 $\pm$ 0.00	2.50 $\pm$ 0.01	0.68 $\pm$ 0.03	0.27 $\pm$ 0.01
Min Bound	2.55	2.55	0.65	0.31
Median Bound	2.55 $\pm$ 0.00	2.59 $\pm$ 0.01	1.15 $\pm$ 0.10	0.40 $\pm$ 0.02
Max Bound	2.55	2.70	1.77	0.61
Average Accuracy	95.90 $\pm$ 0.46	90.15 $\pm$ 0.95	71.23 $\pm$ 2.40	84.90 $\pm$ 1.43

Table 7. Comparison of generative models for sampling parameter candidates on LaMP-2. The diffusion model is the only method that improves upon both the zero-shot baseline and SGD fine-tuning.

Method	Zero-Shot	SGD	Random	GMM	Diffusion
Accuracy	61.68%	63.25%	0.00%	60.29%	63.74%

Table 8. STEEL on CUB-C: certificate metrics across severity levels (16-shot, 5-way; 200 episodes). The non-vacuous ratio is preserved at 100% across all severities, while the bound widens gracefully in proportion to empirical-risk degradation.

	Clean	Mild	Moderate	Severe
Non-Vacuous Ratio	100%	100%	100%	100%
Min Bound	0.28	0.30	0.30	0.30
Median Bound	0.35	0.37	0.38	0.45
Max Bound	0.43	0.52	0.61	0.80
Avg. Accuracy	88.60%	86.70%	84.30%	76.95%

Table 9. MAML vs. STEEL on CUB-C (16-shot, 5-way). MAML provides zero non-vacuous certificates at every severity. Its accuracy advantage over STEEL is at most ~ 1 pp under mild shift, vanishes under moderate shift, and reverses under severe shift.

Severity	Metric	MAML	STEEL
Mild	Non-Vacuous Ratio	0%	100%
	Median Bound	2.63	0.37
	Avg. Accuracy	87.82%	86.70%
Moderate	Non-Vacuous Ratio	0%	100%
	Median Bound	2.65	0.38
	Avg. Accuracy	84.72%	84.30%
Severe	Non-Vacuous Ratio	0%	100%
	Median Bound	2.69	0.45
	Avg. Accuracy	76.47%	76.95%